

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
7	(a)	(i)	Circuit sends current that can be varied through lamp (1) Accept series variable resistor correctly drawn, variable potential divider, or any recognisable symbol for variable (dc) power supply. Ammeter and voltmeter correctly connected and shown by correct symbols (1)	2			2		2
		(ii)	Choose 200 mA as better resolution (accept precision, accuracy) Accept 0.16 A is closer to maximum of range for that meter		1		1		1
		(iii)	More points needed [for lower voltages] (1) accept smaller intervals or more readings pd range needs <u>extending</u> [beyond 2 V (to 3 V)] (1) Don't accept take evenly spaced readings or take repeat readings or equivalent			2	2		2
	(b)	(i)	5.0 [Ω] or $\frac{0.50}{0.10}$ [Ω] or by implication (1) 13.8 [Ω] or $\frac{2.00}{0.145}$ [Ω] or by implication (1) 2.8 (1) Accept correct answer expressed as a fraction 14.2 [Ω] or 14.3 [Ω] are from graph mis-readings deduct 1 mark only		3		3	3	
		(ii)	Marking points M1 [Free] electrons make collisions M2with ions or atoms or lattice (accept particles) M3 Increasing the pd increases the current / drift velocity / energy of electrons M4 Electrons hit ions harder or transfer more energy [or equivalent] when I [or V] larger M5 Increased vibration [amplitude] [of ions/atoms] M6 Therefore temperature increases M7 Collisions more frequent or more probable [when T larger] M8 Smaller current / drift velocity [for a given pd] M9 Resistance increases in context						

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			5-6 marks 7 – 9 of M1 – M9 are present <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks 4 - 6 of M1 – M9 are present <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks 1 - 3 of M1 – M9 are present <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>	6			6		
	(c)		No marks for unsupported 'Spend' or 'Don't spend'. Worth following up because: Any $2 \times (1)$ from: - Use of superconductors avoids energy wastage (accept reduced resistance to zero) or equivalent or specific example pointing out advantage - Room temperature superconductivity means no cooling costs or makes systems simpler or cheaper or equivalent - Advancing scientific knowledge But original claim very likely to be wrong, so hard to defend spending more (1)			3	3		
			Question 7 total	8	4	5	17	3	5